



05) Linear Matrix Inequalities (LMI)

Lyapunov Equation: $A^T P + PA + Q = 0$

Ricatti Equation: $A^T P + PA - PBR^{-1}B^T P + Q = 0$

Lyapunov Equation:

Continuous time:

$$\dot{x} = Ax$$

Lyapunov equation:

$$A^T P + PA + Q = 0$$

Discrete time:

$$x[k+1] = Ax[k]$$

Lyapunov equation:

$$A^T P A - A + Q = 0$$

When there exist set $P > 0$, $Q > 0$, the control system is asymptotically stable.

($x^T P x$ is a Lyapunov function)

Lyapunov Equation (continuous):

Continuous time:

$$\dot{x} = Ax$$

Lyapunov equation:

$$A^T P + PA + Q = 0$$

$$V(x) = x^T P x$$

$$P > 0$$

$$\dot{V}(x) = x^T (A^T P + PA) x$$

$$= -x^T Q x$$

$$Q > 0$$

$V > 0$ and $\dot{V} < 0$ hold for all $x \Rightarrow V$ goes to zero asymptotically.

Lyapunov Inequality (continuous):

Continuous time:

$$\dot{x} = Ax$$

Lyapunov inequality:

$$A^T P + PA + Q < 0$$

$$V(x) = x^T P x$$

$$P > 0$$

$$\dot{V}(x) = x^T (A^T P + PA) x$$

$$< -x^T Q x$$

$$Q > 0$$

$V > 0$ and $\dot{V} < 0$ hold for all $x \Rightarrow V$ goes to zero asymptotically.

Linear Matrix inequalities (LMIs)

Matlab :

1) Initialisation of LMI's environment:

- `setlmi([ ])`

↳ role: Clear the memory and opens up empty space to begin building a new system of matrix inequalities.

2) Declaration of variables (Unknowns):

- `[P, m, sP] = lmiivar(type, structure)`

↳ role: Creates the matrix variable that solver will have to find

inputs: - the type: it is an integer which tells the type of the matrix.

- the structure: it tells the dimension of the matrix and the number of blocks that make it up.

outputs: - P: an internal identifier (a number) that allow Matlab to recognize this variable in subsequent calculations.

- m and sP: the number of internal scalar variables and the structure.

3) Construction of Inequalities (LMI)

- `S1 = newlmi`

↳ role: Creates a new inequality (which will be written in the form $\text{left} < \text{Right}$)

input: none

output: S1: the identifier of this inequality)

- `lmiterm(termID, left-matrix, right-matrix, flag)`

↳ role: it adds the mathematical terms one by one in the inequality S1.

inputs: - [LMI-n°1, block-row1, block-column1, variable-P]

- left-matrix: left from matrix P

- right-matrix: right _____

- `LMI_s = getlmi`

↳ role: "closes" the construction and compiles all the equations.

output: LMI_s: an internal variable that contains the entire system ready to be sent to the solver.

4) Solver Solution.

- `[output1, output2] = mincx(LMI_s, c)`

↳ role: it seeks the value of P that satisfy the inequalities, while trying to minimize an objective (defined by c).

inputs: - LTI: the system
- c: a weighting vector

outputs: - cost: the reached optimal score
- xopt: vector with all the values found.

5) Extraction and Verification

- $P_1 = \text{dec2mat}(LTI, x_{\text{opt}}, P)$
↳ role: translate the raw solution from the solver to a readable matrix.
- $\text{eig}(ans)$
↳ role: gives the eigenvalues.

Lyapunov Equation (Discrete):

Discrete Time:

$$x[k+1] = A x[k]$$

Lyapunov equation:

$$A^T P A - P + Q = 0$$

$$V(x) = x(k)^T P x(k) \quad P > 0$$

$$\begin{aligned} \Delta V &= x(k+1)^T P x(k+1) - x(k)^T P x(k) \\ &= x(k)^T (A^T P A - P) x(k) \\ &= - x(k)^T Q x(k) \quad Q > 0 \end{aligned}$$

$V > 0$ and $\Delta V < 0$ hold for all $x \Rightarrow V$ goes to zero asymptotically.

Lyapunov Inequality (Discrete):

Discrete Time:

$$x[k+1] = A x[k]$$

Lyapunov inequality:

$$A^T P A - P + Q < 0$$

$$V(x) = x(k)^T P x(k) \quad P > 0$$

$$\begin{aligned} \Delta V &= x(k+1)^T P x(k+1) - x(k)^T P x(k) \\ &= x(k)^T (A^T P A - P) x(k) \\ &< - x(k)^T Q x(k) \quad Q > 0 \end{aligned}$$

$V > 0$ and $\Delta V < 0$ hold for all $x \Rightarrow V$ goes to zero asymptotically.

When we obtain P and $Q > 0 \Rightarrow$ System stable.

LMI's, Continuous time:

$$\dot{x} = Ax + Bu$$

$$\begin{pmatrix} A^T P + PA + C^T C & PB \\ B^T P & -\gamma^2 I \end{pmatrix} < 0$$

$$\Rightarrow \begin{pmatrix} x^T & u^T \end{pmatrix} \begin{pmatrix} A^T P + PA + C^T C & PB \\ B^T P & -\gamma^2 I \end{pmatrix} \begin{pmatrix} x \\ u \end{pmatrix} < 0$$

$$\dot{V}(x) + y(t)^2 - \gamma^2 u(t)^2 < 0$$

$$\sup \frac{\|y\|_2}{\|u\|_2} < \gamma$$

, u : input
 y : output
 γ : level of degradation

Design gain K based on H_∞ norm.

$$\dot{x} = Ax + Bu + B_2 w$$

$$y = Cx$$

Find $u = -Kx$ which minimize $\sup \frac{\|y\|_2}{\|w\|_2}$

Feedback controller is given to minimize the effect of the disturbance w to the control output.

Lyapunov inequality (discrete) with feedback $u = -Kx$

$$A^T P A - P < 0$$

feedback $(A - BK)^T P (A - BK) - P < 0$

$$X = P^{-1}$$

$$X (A - BK)^T X^{-1} (A - BK) X - X < 0$$

$$X(A - BK)^T X^{-1} (AX - BKX) - X < 0$$

$$(X^T A^T - X^T K^T B^T) X^{-1} (AX - BKX) - X < 0$$

$$Z = KX$$

$$(X^T A^T - Z^T B^T) X^{-1} (AX - BZ) - X < 0$$

$$(AX - BZ)^T X^{-1} (AX - BZ) - X < 0$$

schur.
$$\begin{bmatrix} -X & (AX - BZ)^T \\ AX - BZ & -X^{-1} \end{bmatrix} < 0$$

$$\Leftrightarrow \begin{bmatrix} -P & (AP - BZ)^T \\ AP - BZ & -P \end{bmatrix} < 0$$

